Shivashankar S Menon

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Work Experience

Founding Engineer - AI @ OnFinance AI Sep 2023 - Mar 2024

- Key areas: **LLMs as orchestrators for retrieval**, **LLM-powered code generation**, optimal knowledge representation of **narrow-domain data**, and **knowledge graphs**.
- Created end-to-end pipelines for data pre-processing, ingestion, retrieval, post-processing, and indexing methods for RAG algorithms. Created Conversational AI product demos for clients like NSE. Demos are later converted into product deployment at scale. Coordinated with the DevOps team to improve the efficiency of integration of Gen AI applications by 33%.

ML Engineer Intern @ American Express Jul 2022 - Dec 2022

Member of R&D team for proprietary xgboost algorithms: helped develop Distributed GPU Learning modes. New modes trained models 20x faster and readied them 5x faster than then-most widely-used mode by their ML engineers. Modes added to company algorithm as official feature update (v3.0; Nov 2022)

R&D Intern @ IIRS Dehradun Jun 2021 - Jul 2021

Implemented a satellite image segmentation model based on the U-Net Architecture using Keras. Analyzed the structure of the Michigan Microwave Canopy Scattering ML Models for simulating microwave back-scattering across forest canopies.

Independent Pursuits

Civil Services (UPSC) Preparation 2024 - 2026

- Self-directed preparation for India's Civil Services (UPSC) examination. I built a holistic understanding of current and historical socio-economic issues, the digital divide, and the scope for AI across Indian industries. Extensive research into public policy and governance sharpened how I analyse complex, real-world problems.
- **Research & Analysis** – synthesised large volumes of information across Economics, Public Administration, and International Relations.
- **Policy & Writing** – broke down complex real-world issues and drafted analytical essays and reports under time constraints.
- **Discipline & Rigor** – sustained intensive daily study routines reinforcing focus, self-management, and high-pressure endurance.

Projects

Accessible Wheelchair Routing Publication Link

Created **two datasets of vibration data** gathered from manual/powered wheelchairs on fifteen surface types. Implemented a surface classification model on the data using **adaptive activation functions**; extended said framework to the powered wheelchair (PW) dataset via **transfer learning** with 98.8% accuracy. **Models outperformed other SOTA models by 3%** with **40% lesser train-time**.

Foreground Emissions Modeling for Astronomy Publication Link

Parsed, analyzed Deep-Sky Observation data gathered from Galaxy Evolution Explorer (GALEX) satellite's spacecraft state files. Derived empirical model for foreground emissions involving **Robust Regression** concepts. Generalized the model to Deep and Medium-Sky Observations.

Education

2019 - 2023	B.E. CS (with DS Minor) at **BITS Pilani, Goa Campus**	(CGPA: 9.15/10)
2018 - 2019	Class 12th CBSE	(92.8%)
2016 - 2017	Class 10th CBSE	(CGPA: 10/10)

Publications

N. Narayanan, S.S. Menon et. al. (Jan. 1, 2023a). "A model of foreground emission in UV using GALEX deep observations". In: *Advances in Space Research* 71.1, pp. 1059–1073. ISSN: 0273-1177. DOI: 10.1016/j.asr.2022.07.086. URL: https://www.sciencedirect.com/science/article/pii/S0273117722007542.

– (May 30, 2023b). "Validation of Foreground Emission Model using GALEX Medium Observations". In: *Research in Astronomy and Astrophysics* 23.6. Publisher: National Astromonical Observatories, CAS and IOP Publishing, p. 065021. ISSN: 1674-4527. DOI: 10.1088/1674-4527/acd09f. URL: https://doi.org/10.1088/1674-4527/acd09f.

V. Misra, S.S. Menon et. al. (Aug. 24, 2024). "AdaGen: Adaptive Generalized Knowledge Transfer Framework for Sensor-Based Surface Classification for Wheelchair Routing". In: *SN Computer Science* 5.7, p. 820. ISSN: 2661-8907. DOI: 10.1007/s42979-024-03181-w. URL: https://doi.org/10.1007/s42979-024-03181-w.

Skills

ML/DL/AI	Numpy, Pandas, Scikit-Learn, TensorFlow, PyTorch, Claude Code
Programming	OOP, DBMS, DSA
Languages	Python, LaTeX
Policy & Analysis	Public Policy, Economics, Public Administration, International Relations, Good Governance, Research & Synthesis, Analytical Writing